Maastricht University
School of Business and Economics

PETRA BÉKÉSI

INTRODUCTION TO BUSINESS ANALYTICS

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ENHANCED DECISION MAKING BASED ON RELIABLE FORECASTS

WEATHER PREDICTION FOR HORNS REV 1

ANNA GUSTAVA MITCHELL

I6386985

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Introduction

“If you stand on the western coast of Jutland in Denmark and look at the horizon, you can just discern one of the world’s largest offshore wind farms […] There, at Horns Rev, the wind conditions are ideal for wind power generation, enabling the wind farm’s 80 turbines to generate electricity for around 150,000 private households in Denmark.” (Vattenfall, 2024) Offshore wind farms are especially important for generating green energy since they are able to produce almost twice the energy of onshore wind farms. Due to the high average in wind speed and the possibility to essentially operate around the clock, being able to predict the weather conditions can help immensely with energy optimization. Moreover, the severe sea weather and poor accessibility cause considerable difficulties in maintaining offshore wind turbines (Xiaolin, et al., 2020), highlighting the significance of predicting the upcoming weather. This is possible by combining data repositories with advanced analytics techniques to create an accurate prediction model that would for example be able to predict where and with which speed the wind would be coming from tomorrow. Consequently, Horns Rev would be able to position the wind turbines in a way, that would minimize the wake effect and optimize the energy production.

This business proposal will explain how Horns Rev 1 can utilize a weather prediction model to promote the growth and efficiency of the wind farm.

Company Description

Horns Rev 1 is a Dutch Offshore Wind Farm renewable energy project in the North Sea and was the first large-scale wind farm in the world, originally entered into commercial operation in December of 2002. The plant run by Vattenfall, consists of 80 turbines, each with an output capacity of 2MW, adding up to a total maximum capacity of 160MW. (GlobalData, 2024) With a height of 70m, the wind turbines are usually met with average wind speeds of 9 to 10m/s “which creates ideal conditions for wind power generation.” (Vattenfall, 2024)

Vattenfall is an energy company, operating in Denmark and many other European cities for over 100 years. With the help of approximately 19,000 employees, they have been able to supply nearly 8 million households with electricity, further the advancement towards the freedom from fossil fuels, and work to reach net-zero by 2040. (Vattenfall, 2024)

Business Opportunities

Horns Rev 1 has a nameplate capacity of 160MW however the capacity factor is 40-45% so there is a possibility to produce more energy. With the help of this predictive model, Horns Rev would be able to increase their capacity factor by adjusting their operations. (Campbell, 2019)

The Nacelle, the housing around the generating components of a wind turbine, is able to turn the turbines into a specific direction. With the prediction of the wind direction, the turbines in Horns Rev 1 can be positioned in a way that would maximize the energy earnings. The continuous winds that Horns Rev is exposed to in its position in the North Sea, set it apart from wind farms inland. Combining this advantage with a predictive model for wind speed and direction, it would be possible to estimate the direction it would be most beneficial to turn into,

based on the wind speed. This can increase the capacity factor of the offshore wind farm, optimizing its output.

Data Analytics Technique: Multiple Linear Regression

Multiple linear regression is a statistical analysis tool that is able to use one dependent (response) variable and multiple independent variables, also known as explanatory variables, to better visualize the influence of the predictors on the response variable. The general form of this is given in an equation, which is presented below:

Equation 1: $$f(x) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

In the equation, every x represents an independent variable and every w represent the corresponding weight – the magnitude of each weight roughly corresponds to the significance of the variable in predicting the explanatory variable. The equation is the outcome of multiple linear regression finding a straight line that best describes the connection between the independent and dependent variables as well as the equation that best fits the data. Multiple linear regression uses historic data from the variables, for example weather elements like wind speed and direction, to create the line equation. The data is analyzed to find a measure for the correlation between the different variables. We receive the equation by running a regression analysis in excel. The regression gives us values like the "R^2" and the "adjusted R^2" which allow us to evaluate the model's accuracy at first glance. Furthermore it is also possible to evaluate the correlation by interpreting the p-value or calculating the absolute average error.

In multiple linear regression, the "regression coefficient" or "R^2" is a statistical value, measured on a scale from 0 to 1, that numerically describes how the variability in the dependent variable can be explained through the independent variables. The closer R^2 is to 1, the more the independent variables explain the variability of the dependent variable.

The "adjusted R^2" is a further tool to evaluate the significance of the independent variables. R squared increases with each variable added that explains the dependent variable. If the values are too similar it can lead to overfitting in the model, resulting in an "inflated" R^2. That is why the adjusted r squared is used to double check. It is "adjusted" to the number of predictors and therefore changes, depending on how significant the variables are to the model. If the added variable does not improve the model, the adjusted r squared decreases. (The Investopedia Team, 2024)

The p-value is an indicator of whether the relationships between the variables are statistically significant. The p-value tests the null hypothesis H_0:

There is no correlation between the response and explanatory variables.

If the P-Value is low ($p<0.05$), it indicates that the observed effect is unlikely to be a result of chance, suggesting a relationship between the variables. This leads to rejecting the null hypothesis. On the other hand, if the p-value is higher than the significance level of 0.05, the null hypothesis cannot be rejected, as there is a chance any observed differences are due to random variation.

Lastly, the model can be evaluated by the Average Absolute Error. It is the absolute average difference between the holdout data of the city we want to predict and the predicted data. By applying the equation the regression analysis creates and calculating the difference, we can see on average how far off the predicted values are from the actual values. This gives us a numerical value to assess the model's accuracy and allows us to evaluate if it overfits or not.

The outlined statistical metrics are ways to evaluate the quality of multiple linear regression models and indicate how well the explanatory values describe the response variable. Furthermore, multiple linear regression is a fairly comprehensible method of predicting a linear correlation between different variables. Conversely, this model cannot predict significant values, if the correlation is not linear or the data has prominent outliers.

The proceeding section will explain in detail how a multiple linear regression model can be used to create a weather prediction model that is beneficial to this company. The data-driven-decision making will allow Horns Rev 1 to improve their energy output.

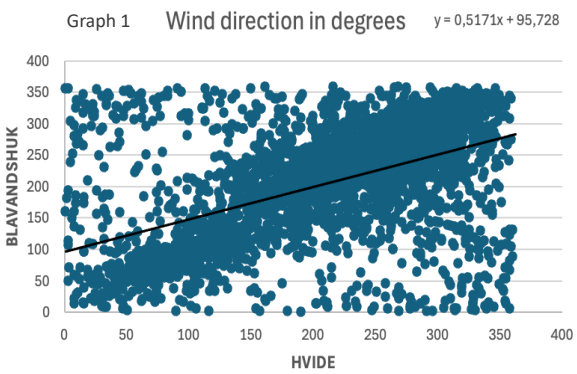
Model Building

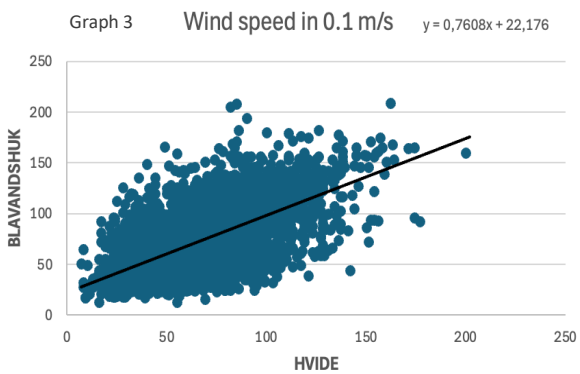
This model uses historical data from closely located cities, in order to provide an accurate prediction of wind speed and direction. It is most profitable to have detailed information on these factors because they affect the production of wind energy the most. As the target city, the weather station Blavandshuk Fyr was chosen due to its distance from the offshore wind farm Horns Rev 1. These locations are 7km apart and the predicted data for Blavandshuk Fyr is therefore substantial for Horns Rev 1. The other two target cities are Hvide Sande and Vester Vedsted, located north and south of the target city. This allows a precise prediction of the weather elements by accounting for typical wind patterns and seasonal variation.

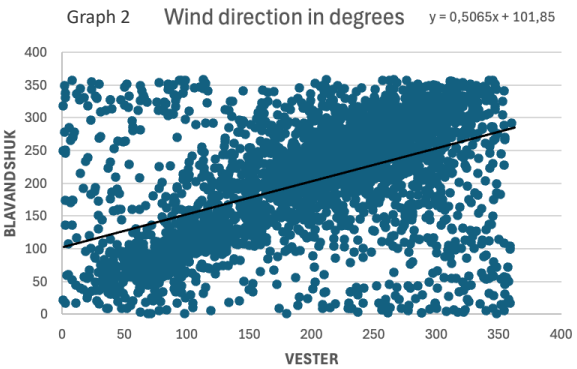
The European Climate Assessment & Database (ECAD) provides the historical data needed to design this model. The weather stations relevant to this project only reported data from the beginning of 2011, however the data is sufficient to create a reliable predictive model. As previously mentioned, this business proposal will be focusing on wind speed and wind direction, so the required data was downloaded from ECAD.

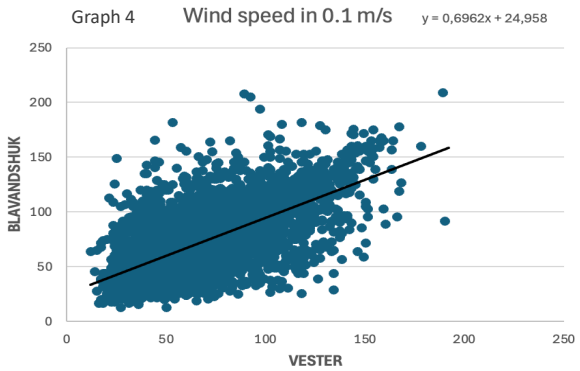
After the data is acquired, it has to be prepared. For the training data, the data that in the end would build the model, the timeframe from 2011 until 2018 was decided upon. The last five years, called the holdout data, were put to the side for later use. This is the data the model will be tested on to see how accurate it is. The data was then prepared, which included cleaning out the missing values. Since we want to predict the data at the wind farm a day ahead, the training data of our target city's starting date will be t and our other cities date t-1. For this, the first value of Blavandshuk Fyr is removed and the last value of Hvide Sande and Vester Vedsted. That means that the data now starts on January 3rd and 2nd for the associated cities.

To clearly see the correlation between the different data sets, the cities are plotted against each other in a scatter plot. This visualizes the likeness of the elements in the different cities. On the left side are Hvide Sande and Vester Vedsted plotted against Blavandshuk Fyr in with respect to wind direction and on the right is shown the variable wind speed.









The graphs have a regression line with the equation stated at the top. The wind direction graphs have a lot of outliers, and the plotted data only scarcely follows the trend line. The wind speed's graphs are a bit better, with the data being gathered in a bubble like shape around the trend line, suggesting a closer correlation. These graphs provide a first impression, if the independent variables would have an adequate predictive power. It is important however to keep in mind that these cities are all along the coast which means they are subjected to coastal weather that can be very unpredictable. For these reasons, I decided to choose multiple linear regression, run with the data of the three cities, since additional variables improve predictive accuracy.

The following section will concentrate on the multiple linear regression of wind direction. The results of the procedure are shown in the table below:

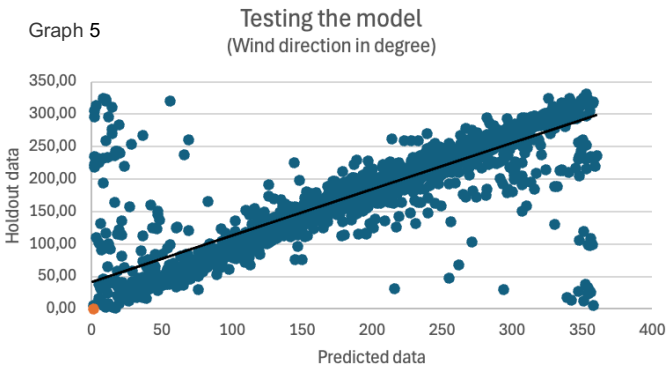
SUMMARY OUTPUT								
					Average absolute error:		30,29	
Regression Statistics								
Multiple R	0,8446652							
R Square	0,7134592							
Adjusted R Square	0,7131816							
Standard Error	49,438683							
Observations	2067							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	2	12561057,45	6280528,7	2569,5816	0			
Residual	2064	5044794,538	2444,1834					
Total	2066	17605851,99						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*
Intercept	17,171202	2,875734479	5,9710664	2,768E-09	11,531558	22,810845	11,531558	22,810845
X Variable H	0,2822746	0,01639955	17,21234	3,79E-62	0,2501132	0,314436	0,2501132	0,314436
X Variable V	0,650137	0,017021495	38,195056	6,6E-242	0,6167559	0,6835181	0,6167559	0,6835181

Table 1

The blue highlighted cells are the ones we will be looking at to evaluate how well our model will predict the wind direction in our target city a day prior. The R squared value of 0.71 indicates that 71% of the variance in the wind direction of Blavandshuk Fyr can be explained by the variance of the wind direction in the other two cities. This is a visibly higher percentage than the graphs would have indicated. This shows how multiple linear regression is able to enhance predictive accuracy. Comparing the R^2 to the adjusted R^2, there is almost no difference, with the adjusted r squared being smaller by a fraction. This indicates that our explanatory variables provide valuable insights for the wind direction in Blavandshuk Fyr.

To assess the accuracy of our model, the holdout data (01.01.2019 until 30.06.2024) is needed. This was prepared as well in the same manner as the training data. Going back to the holdout data, we create an equation using the purple highlighted values. These are substituted into the previously mentioned equation: $f(x) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$ This gives us predicted values one day in advance.

To evaluate the accuracy of the predicted values, the methods mentioned in the technique description will be used. To start, one can look at the p-values. The values highlighted in pink are all approaching zero ($p<0.05$), meaning that the chosen cities provide significant information on the wind direction in Blavandshuk Fyr. Next, the average distance is calculated. This is done by subtracting the predicted values from the actual values. However it is important to calculate the absolute difference, so the different values don't cancel each other out. The average of the difference is highlighted on the top right. An average absolute error of 30.3 means that the predicted wind direction differs from the actual direction by about 30°, which should not be ignored however it will not affect the prediction too much. The error seems to be higher than expected however it is important to keep in mind that the model is for wind direction and if the wind switches from 360° (north wind) to 5° (north), over the course of a day, the average will be thrown off a bit. The average error helps us determine if the model overfits. Since an error of 30° is not excessively high, we can conclude that the model doesn't overfit and therefore hold significant predictive power. The following graph visually demonstrates how well the model will predict the data:



It is clearly visible that the predicted data and the holdout data have a correlation with most of the values being centered along the line of regression. Only a few outliers are seen around the edges, this is due to the previously explained challenge. To further review this prediction, a baseline model is used. This is created by simply predicting the data with the average of the wind direction in the two explanatory cities a day prior and calculating the average absolute error the same way as above. This gives us a value or 78.3° which means that using a baseline model, the average absolute error would be about 78° which shows that our predictive model has an accuracy twice as good. Furthermore, it is beneficial to be able to predict the wind

speed at the target location in advance. Combining the two models for wind direction and speed allows the wind farm to position the wind turbines in the most beneficial direction, ready to capture the maximum wind energy.

The following section will evaluate the wind speed predictive model:

SUMMARY OUTPUT								
Regression Statistics					Average absolute error:		26,27	
Multiple R	0,5177134							
R Square	0,2680272							
Adjusted R Square	0,2673179							
Standard Error	25,848821							
Observations	2067							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	2	504981,1067	252490,55	377,88848	1,44E-140			
Residual	2064	1379085,439	668,16155					
Total	2066	1884066,546						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*
Intercept	32,981595	1,56163211	21,119952	8,443E-90	29,919056	36,044133	29,919056	36,044133
X Variable H	0,359308	0,034809137	10,322231	2,166E-24	0,2910433	0,4275726	0,2910433	0,4275726
X Variable V	0,2229217	0,038709686	5,7588091	9,741E-09	0,1470076	0,2988358	0,1470076	0,2988358

Table 2

The preparation for this model as well as the time frame of the training data (2011-2018) and for the holdout data (2019-2024) is the exact same as the one for wind direction, so I will not go into detail about the design process of the model.

Looking at the table above, it is clearly visible that the r squared is much lower than the previous one. This means that 27% of the variance in the wind speed of Blavandshuk Fyr can be explained by the variance of the wind speed in the other two cities. The value is lower than before, suggesting that the predictive values will be less accurate. However the adjusted r squared is still very similar in value to the r squared which means that the variables do explain the predictive values.

To properly understand the value the model holds, the average absolute error is calculated again as explained for the other model. The average absolute error is 26.27. Since the values in this data set are given in 0.1 m/s, an error of 26.27 means that the predicted values will vary by about 2.6 m/s from the actual data which is a minimal amount. Comparing this to a baseline model designed the same way as for wind direction, the average error would be 27.9 so the values are off by 2.8 m/s. This suggests that our predictive model is again better than the baseline model. Consequently, even though the R^2 was lower, the model still holds meaning for predicting wind speed at the target location.

Conclusion

To conclude the evaluation of the models, I believe it is clear that the models have significant value in predicting the data one day ahead of time. Therefore these weather prediction models would improve the data-driven-decision making processes at Horns Rev 1. There are many predictive models however, as the comparison to the baseline model showed, however this model is better at predicting the wind reliably and accurately. Hence, it would be possible to

predict from where the wind would be coming tomorrow and at which speed. The wind turbines can be programmed ahead of time so that they are ready to generate energy when the wind is at its maximum. Coastal weather does not always have consistent wind patterns however an accurate prediction can enhance the overall efficiency. Furthermore, a problem with windfarms is that a continuous production is not always possible because of cut-in (3-4 m/s) and cut-out (>25 m/s) wind speeds. The wind turbines can only operate in between the two margins. The predictive model can therefore predict when these low or excessive winds would occur, also minimizing the chance of potential damage due to winds higher than 25 m/s. (Block, 2024)

An additional factor, which influence can be minimized with the help of predicted wind direction, is the wake effect. The wake effect is the turbulent airflow after a wind turbine because of the disrupted wind flow, consequently the wind loses speed and energy. By aligning the wind turbines axes parallel to the primary wind flow, the turbulences behind them can be reduced, as well as the effect on other turbines. This is possible through the model's prediction of wind direction for the upcoming day. Moreover, the turbines could be programmed to turn in advance, leaving only corrections the day of, increasing operational efficiency.

To validate the model and asses the reliability, multiple tests were run. Firstly, the holdout data makes sure that the model actually works and is able to predict applicable values. Secondly, the average absolute error guarantees that the model does not overfit, which means that the model is excessively complex, looking for any sort of pattern in the training data. This can be a huge problem since the model would then be unapplicable in real-life situations. However due to the generally low average absolute error, it is clear to say that the model does not overfit.

Ultimately, I can say that the models for predicting wind speed and direction are of substantial value to Horns Rev 1, due to factors including the ones previously stated. The goal of the predictive model is to allow Horns Rev 1 to increase their capacity factor of 40-45%. Even though Horns Rev 1 is already at an advantage since offshore wind farms typically have a higher capacity factor, ideally the factor would be increased to 50-60% of the nameplate capacity. This would be an increase in energy production of about 20-30MW, allowing them to supply almost 17,000 more households in Denmark. (Marsh, 2024)

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